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Thermal buckling analysis of porous functionally graded nanocomposite beams reinforced by graphene platelets using Generalized differential quadrature method

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A B S T R A C T

In this paper, thermal buckling of functionally graded (FG) porous nanocomposite beams subjected to a thermal gradient are studied by generalized differential quadrature method (GDQM). We consider three different types of nanofillers dispersion patterns and porosity distributions. Materials parameters vary along the thickness direction. Under Gaussian random field (GRF) scheme, the mechanical properties of closed- cell cellular solids are used. Thereby, the variation of Poisson's ratio as well as the relationship between porosity coefficient and mass density are determined. The elastic modulus of nanocomposite is obtained by applying Halpin-Tsai micromechanics model. In the course of this work, the accuracy and efficiency of the (GDQM) are validated. We studied the effects of weight fraction, dispersion pattern, geometry, and size of graphene platelets (GPLs), as well as porosity distribution, porosity coefficient, slenderness ratio and metal matrix on the thermal buckling of the nanocomposite beam. Our findings, contrary to what was expected, are somewhat surprising. According to our results, the graphene platelets (GPLs') performance is affected strongly by their geometry.

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1. Introduction

The first attempts to apply porous metals for engineering application date back to the beginning of the 20th century.

Here's what have made the porous materials attractive for researches: lightness, electrical conductivity, energy absorption, and thermal management [1–3].

In doing so, porous materials have become a topic of great interest to researchers. For instance, the nonlinear vibration of a higher-order refined four-variable nanoplate made of nanoporous materials was analyzed in [4]. The nonlinear frequency was obtained by using a homotopy perturbation method and a novel Hamilton approach. Meanwhile the small-scale effects were incorporated by the Eringe's nonlocal elasticity theory. In another case, Akbaş [5] reported the forced vibration analysis of functionally graded porous deep beams. The studied beam was under a harmonic external distributed load. The porous biomaterials are the trending area of the biological research capturing the attention of many researchers; for example, by employing the Galerkin technique, Sahmani and et al. [6] studied the size-dependent nonlinear secondary resonance of a micro nano-beam made of the nano-porous biomaterials. The mechanical properties of the nano-porous biomaterials were obtained as the function of pore size. Anamagh and Bediz [7] studied free vibration and buckling behavior of functionally graded porous plates reinforced by graphene platelets. They used spectral Chebyshev approach and concluded; buckling strength and vibration behavior depend highly on the dispersion of porosity and nanofiller material along the thickness of the composite. Babaei et al. [8] considered thermal bucking and post-bucking responses of geometrically imperfect FG porous beams based on physical neutral plane. A two-step perturbation technique was used to obtain the thermal buckling and post-buckling responses of FG porous beams. Liu et al. [9] investigated thermal-mechanical coupling buckling analysis of porous functionally graded sandwich beams based on physical neutral plane. The results showed the critical temperature increases greatly with the rise of porosity. Zenkour [10] addressed the thermal buckling of actuated functionally graded piezoelectric porous nanoplates. Eringen's nonlocal elasticity theory with a higher-order shear deformation theory was used to obtain the analytical solution. Zhenhuan et al. [11] considered accurate nonlinear buckling analysis

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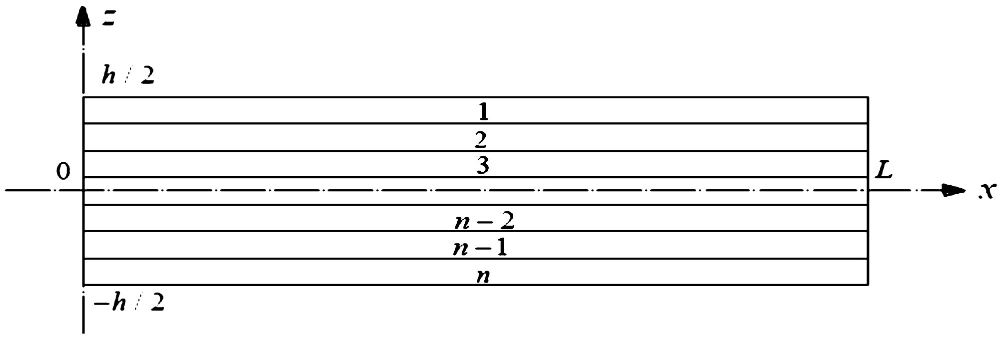

Fig. 1. Configuration of the studied beam.

of a functionally graded porous graphene platelet reinforced composite cylindrical shells under axial compressive load. They concluded the symmetric dispersion pattern is the optimal material distributions for both graphene platelets and porous, and the largest possible weight fraction, specific surface area and average thickness of graphene platelets could induce a better anti-buckling performance for the nanocomposite shell.

Nanofillers added to the porous materials must simultaneously have three features, viz, they should be ultra-lightweight so that, by adding them to materials, the reinforced materials should remain in the lightweight structure category; their introducing should improve the mechanical properties of the reinforced materials; and they should have stable structure in chemical viewpoint. Carbon nanotubes (CNTs) and GPLs [12–14] have exceptionally the three above-mentioned features, so both of them are appropriate candidates as the nanofiller.

To reveal the performance of nanocomposite structures reinforced by CNTs, the numerous studies have been done such as [15–18].

The references [14,19,20] compared the performance of GPLs and CNTs in the modification of the mechanical properties of structures and it can be deduced that GPLs perform even better.

There is a lot of research on the use of DQM method in solving governing equations in FG beams. Recently Patil [21] used DQM to solve the governing differential equation of motion for vibration control of a functionally graded material (FGM) beam using magnetostrictive layers. He obtained the results for various boundary conditions. Lal and Saini [22] employed DQM as a powerful numerical solution to study vibration analysis of FGM circular plates under non-linear temperature. Researchers considered free vibration analysis of saturated porous FG circular plates integrated with piezoelectric actuators [23]. They used DQM for the solution to show the efficiency of this method. Currently Bo et al. [24] used a novel DQM -based geometric mapping scheme to address the C^1-continuity requirements of the in-plane and out-of-plane kinematic quantities of a couple stress-based Mindlin–Reissner functionally graded (FG) plate model.

In spite of many literatures on the mechanical behavior of graphene and its derivation such as [25–27], this set of researches still holds great potential for more studies.

To our knowledge, rare studies to date have tended to consider the effect of nanofillers' patterns geometric characters on their performance. Therefore, it would be of special interest to analyze the effect of porosity and GPLs dispersion patterns - less attended topics - on the mechanical behavior of structures.

The overall goal of this study is to pursue the effect of porosity and GPLs, chiefly their geometric parameters, on the thermal buckling of FG Timoshenko beam. To this end, the solution plan established is based on three steps: first, utilizing the Timoshenko beam theory to describe theoretical formulations, second, deriving governing equations through using Hamilton's principle, and third, applying Generalized differential quadrature method (GDQM) for the discretization of equations ultimately led to the calculation of critical thermal gradient. The unforeseeable results suggest that the geometrical parameters of GPLs have an extraordinary effect on the thermal buckling of the current nanocomposite beam also these beams perform too acceptably.

2. Porosity distributions, GPLs patterns, and mechanical properties

Fig. 1 shows a schematic diagram of a porous multilayer beam with a Cartesian coordinate system which its x-axis is on the mid-plane and z-axis is parallel to the thickness direction. We use the following notation for beam: h, thickness; b, width; L, length.

Three different GPLs dispersion patterns namely A, B, and C are looked carefully at each porosity distribution in Fig. 2. The GPLs volume content ($\mathrm{V_{GPL}}$) changes along the z-axis smoothly and the peak values of V_{GPL} are determined according to the specific porosity distribution. It is supposed that the total amount of GPLs in three different GPLs dispersion patterns is the same and this leads to $s_{i1} \neq s_{2i} \neq s_{3i}$.

It should be noticed that E'_1 and E'_2 are maximum and minimum Young's moduli of the non-uniform porous beams without GPLs, respectively, and E' stands Young's modulus of the beam with uniform porosity distribution.

To determine the mass density $\boldsymbol{\rho}((\mathbf{z}))$, elastic moduli ($\boldsymbol{E}(\boldsymbol{z})$), and thermal expansion coefficient $\boldsymbol{\alpha}(\mathbf{z})$ of the FG porous nanocomposite beams with non-uniform symmetric, non-uniform asymmetric, and uniform porosity distribution, the following relationships are used:

$$\begin{aligned} \boldsymbol{\rho}(\mathbf{z}) &= \boldsymbol{\rho_1}[\mathbf{1} - \boldsymbol{e_m}\lambda(\boldsymbol{z})] \\ \boldsymbol{E}(\boldsymbol{z}) &= \boldsymbol{E_1}[\mathbf{1} - \boldsymbol{e_0}\lambda(\boldsymbol{z})] \\ \boldsymbol{G}(\boldsymbol{z}) &= \boldsymbol{E}(\boldsymbol{z})/[\mathbf{2}(\mathbf{1} + \boldsymbol{\nu}(\boldsymbol{z}))] \end{aligned} \tag{1}$$

and according to [28]:

$$\boldsymbol{\alpha}(\mathbf{z}) = \boldsymbol{\alpha_1} \tag{2}$$

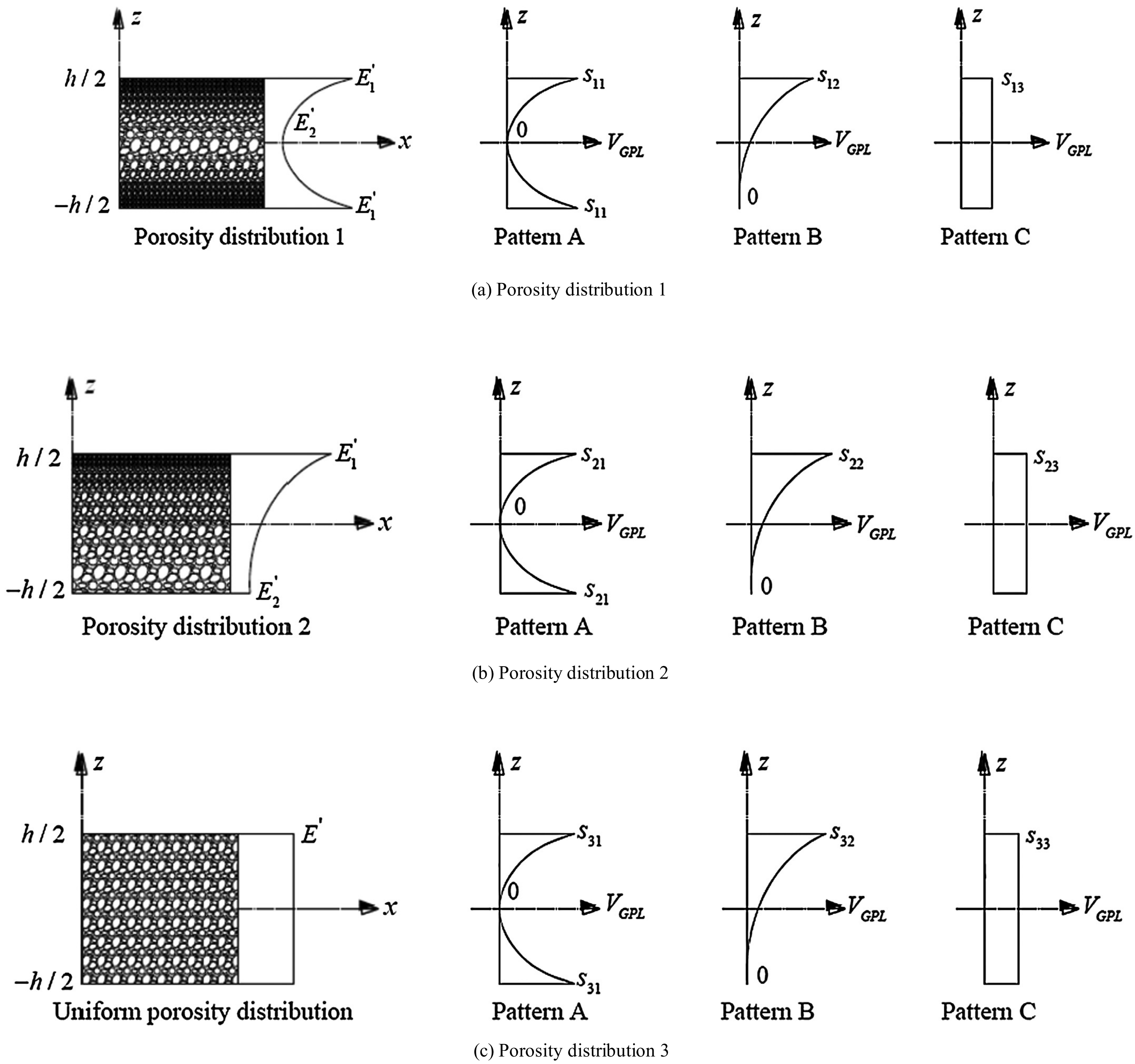

Fig. 2. Porosity distributions and GPL dispersion patterns. (a) Non-uniform porosity distribution 1 with GPLs dispersion patterns. (b) Non-uniform porosity distribution 2 with GPLs dispersion patterns. (c) uniform porosity distribution with GPLs dispersion patterns.

where

$$\lambda(z) = \begin{cases} \cos(\pi z/h) & \textit{Porosity distribution 1} \\ \cos(\pi z/2h + \pi/4) & \textit{Porosity distribution 2} \\ \lambda & \textit{Uniform distribution} \end{cases} \tag{3}$$

where E_1, ρ_1, and α_1 are the maximum values of Young's modulus, mass density, and thermal expansion coefficient of the porous beam, respectively. The porosity coefficient is defined as follows:

$$e_0 = 1 - \frac{E_2'}{E_1'} \tag{4}$$

According to [29], in the Gaussian Random Field (GRF) scheme; the mechanical property of closed-cell cellular solids is expressed as:

$$\frac{E(z)}{E_1} = \left(\frac{\rho(z)/\rho_1 + 0.121}{1.121}\right)^{2.3}, \qquad \left(0.15 < \frac{\rho(z)}{\rho_1} < 1\right) \tag{5}$$

Using Eq (5), the coefficient of mass density e_m is determined as follows:

$$e_m = \frac{1.121\left(1 - \sqrt[2.3]{1 - e_0\lambda(z)}\right)}{\lambda(z)} \tag{6}$$

Also, Poisson's ratio $v\,(z)$ can be calculated based upon the closed-cell GRF scheme [30].

$$\upsilon\,(z) = 0.221p' + \nu_1\left(0.342p'^2 - 1.21p' + 1\right) \tag{7}$$

where ν_1 refers to Poisson's ratio of pure non-porous matrix materials and

$$p' = 1 - \frac{\rho(z)}{\rho_1} = 1.121\left(1 - \sqrt[2.3]{1 - e_0\lambda(z)}\right) \tag{8}$$

Assuming the sameness of the beams total mass with different porosity distribution, λ, the related term in Eq. (3), can be derived as:

$$\lambda = \frac{1}{e_0} - \frac{1}{e_0}\left(\frac{M/h + 0.121}{1.121}\right)^{2.3} \tag{9}$$

where M is identical for all porosity distributions and can be obtained by:

$$M = \int_{-h/2}^{h/2}\left(1 - p'\right)dz \tag{10}$$

In accord to the dispersion patterns depicted in Fig. 2, the volume fraction of GPLs can be shown as (i =1, 2, 3):

$$V_{GPL} = \begin{cases} s_{i1}\left[1 - \cos\left(\pi z/h\right)\right] & \textit{Pattern A} \\ s_{i2}\left[1 - \cos\left(\pi z/h + \pi/4\right)\right] & \textit{Pattern B} \\ s_{i3} & \textit{Pattern C} \end{cases} \tag{11}$$

The weight fraction of GPLs Λ_{GPL} is related to V_{GPL} via:

$$\frac{\Lambda_{GPL}}{\Lambda_{GPL} + \frac{\rho_{GPL}}{\rho_M} - \frac{\rho_{GPL}}{\rho_M}\Lambda_{GPL}} \times \int_{-h/2}^{h/2}\left[1 - e_m\lambda(z)\right]\mathrm{d}z = \int_{-h/2}^{h/2} V_{GPL}\left[1 - e_m\lambda(z)\right]dz \tag{12}$$

To determine the elastic modulus of the non-porous nanocomposites, the Halpin-Tsai micromechanics model [14,31–33] is applied.

$$E_1 = \frac{3}{8}\left(\frac{1 + \xi_L^{GPL}\eta_L^{GPL}V_{GPL}}{1 - \eta_L^{GPL}V_{GPL}}\right)E_M + \frac{5}{8}\left(\frac{1 + \xi_W^{GPL}\eta_W^{GPL}V_{GPL}}{1 - \eta_W^{GPL}V_{GPL}}\right)E_M \tag{13}$$

where

$$\begin{aligned} \xi_L^{GPL} &= \frac{2l_{GPL}}{t_{GPL}} \\ \xi_W^{GPL} &= \frac{2w_{GPL}}{t_{GPL}} \\ \eta_L^{GPL} &= \frac{(E_{GPL}/E_M) - 1}{(E_{GPL}/E_M) + \xi_L^{GPL}} \\ \eta_W^{GPL} &= \frac{(E_{GPL}/E_M) - 1}{(E_{GPL}/E_M) + \xi_W^{GPL}} \end{aligned} \tag{14}$$

where w_{GPL}, l_{GPL}, and t_{GPL} denote the average width, length, and thickness of GPLs, respectively, and E_M is Young's modulus of the metal.

The mass density ρ_1, Poisson's ratio ν_1, and thermal expansion coefficient α_1 of the metal matrix reinforced by GPLs are calculated by the following mixture rule.

$$\begin{aligned} \rho_1 &= \rho_{GPL}V_{GPL} + \rho_M V_M \\ \nu_1 &= \nu_{GPL}V_{GPL} + \nu_M V_M \\ \alpha_1 &= \alpha_{GPL}V_{GPL} + \alpha_M V_{GPL} \end{aligned} \tag{15}$$

in which $\rho_{GPL}, \nu_{GPL}, \alpha_{GPL}$, and V_{GPL} are the mass density, Poisson's ratio, thermal expansion coefficient, and volume fraction of GPLs, respectively. In addition, ρ_M, ν_M, α_M, and $V_M = (1 - V_{GPL})$ are corresponding properties for metals.

3. Theory and formulations

3.1. Equations of motion

Based on the Timoshenko beam theory, the formulas of the displacement field of the beam are expressed in the x and z directions:

$$\boldsymbol{W}\left(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{t}\right)=\boldsymbol{w}_{\boldsymbol{o}}\left(\boldsymbol{x}, \boldsymbol{t}\right), \ \boldsymbol{U}\left(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{t}\right)=\boldsymbol{u}_{\boldsymbol{o}}\left(\boldsymbol{x}, \boldsymbol{t}\right)+\boldsymbol{z}\boldsymbol{\phi}\left(\boldsymbol{x}, \boldsymbol{t}\right) \tag{16}$$

Where $u_0(x,t)$ and $w_0(x,t)$ stand the axial and transverse displacement components on the mid-plane (z =0), respectively. $\phi(x,t)$ represents the transverse normal rotation about the y-axis and t is time. The linear strain-displacement relationship provides the axial and transverse shear strains as:

$$\begin{aligned}
&\boldsymbol{\varepsilon}_{\boldsymbol{x}}=\frac{\partial \boldsymbol{u}_{\mathbf{0}}}{\partial \boldsymbol{x}}+\boldsymbol{z}\frac{\partial \boldsymbol{\phi}}{\partial \boldsymbol{x}}\\
&\boldsymbol{\gamma}_{\boldsymbol{xz}}=\frac{\partial \boldsymbol{w}_{\mathbf{0}}}{\partial \boldsymbol{x}}+\boldsymbol{\phi}
\end{aligned} \tag{17}$$

The linear stress-strain constitutive law can represent the forms of the normal and transverse stresses.

$$\begin{aligned}
&\boldsymbol{\sigma}_{\boldsymbol{x}}=\boldsymbol{Q}_{\mathbf{11}}\left(\boldsymbol{z}\right)\left[\boldsymbol{\varepsilon}_{\boldsymbol{x}}\left(\boldsymbol{x}\right)-\boldsymbol{\alpha}\left(\boldsymbol{z}\right)\boldsymbol{\Delta T}\left(\boldsymbol{z}\right)\right]\\
&\boldsymbol{\tau}_{\boldsymbol{xz}}=\boldsymbol{Q}_{\mathbf{55}}\left(\boldsymbol{z}\right)\boldsymbol{\gamma}_{\boldsymbol{xz}}\left(\boldsymbol{x}\right)
\end{aligned} \tag{18}$$

where the elastic elements $Q_{11}(z)$ and $Q_{55}(z)$ are:

$$\begin{aligned}
&\boldsymbol{Q}_{\mathbf{11}}\left(\boldsymbol{z}\right)=\frac{\boldsymbol{E}\left(\boldsymbol{z}\right)}{\mathbf{1}-\boldsymbol{\upsilon}^{\mathbf{2}}(\boldsymbol{z})}\\
&\boldsymbol{Q}_{\mathbf{55}}\left(\boldsymbol{z}\right)=\frac{\boldsymbol{E}\left(\boldsymbol{z}\right)}{\mathbf{2}\left(\mathbf{1}+\boldsymbol{\upsilon}(\boldsymbol{z})\right)}=\boldsymbol{G}(\boldsymbol{z})
\end{aligned} \tag{19}$$

The governing equations of motion are derived from Hamilton's principle.

$$\int_{\mathbf{0}}^{\boldsymbol{t}}\left(-\delta\boldsymbol{\Pi}+\delta\boldsymbol{V}+\delta\boldsymbol{T}\right)\boldsymbol{dt}=\mathbf{0} \tag{20}$$

Where δ signifies the variational symbol.

The strain energy of the porous nanocomposite beam in terms of the strains and stresses is formulated by:

$$\delta\boldsymbol{\Pi}=\frac{\boldsymbol{b}}{\mathbf{2}}\int_{\mathbf{0}}^{\boldsymbol{L}}\int_{-\boldsymbol{h}/\mathbf{2}}^{\boldsymbol{h}/\mathbf{2}}\left(\boldsymbol{\sigma}_{\boldsymbol{x}}\boldsymbol{\varepsilon}_{\boldsymbol{x}}+\boldsymbol{\tau}_{\boldsymbol{xz}}\boldsymbol{\gamma}_{\boldsymbol{xz}}\right)\boldsymbol{dzdx} \tag{21}$$

The potential energy V, arising from the external axial load N_{x0}, and kinetic energy T of the beam are:

$$\delta\boldsymbol{V}=\frac{\boldsymbol{b}}{\mathbf{2}}\int_{\mathbf{0}}^{\boldsymbol{L}}\left[\boldsymbol{N}_{\boldsymbol{x}\mathbf{0}}\left(\frac{\partial\boldsymbol{W}}{\partial\boldsymbol{x}}\right)^{\mathbf{2}}\right]\boldsymbol{dx} \tag{22}$$

$$\delta\boldsymbol{T}=\frac{\boldsymbol{b}}{\mathbf{2}}\int_{\mathbf{0}}^{\boldsymbol{L}}\int_{-\boldsymbol{h}/\mathbf{2}}^{\boldsymbol{h}/\mathbf{2}}\boldsymbol{\rho}(\boldsymbol{z})\left[\left(\frac{\partial\boldsymbol{U}}{\partial\boldsymbol{t}}\right)^{\mathbf{2}}+\left(\frac{\partial\boldsymbol{W}}{\partial\boldsymbol{t}}\right)^{\mathbf{2}}\right]\boldsymbol{dzdx} \tag{23}$$

The following equations of motion are derived through three steps: substituting Eqs. (21)-(23) in Hamilton's principle, integrating through the beam thickness direction, and setting coefficients of δu, δw, and $\delta\varphi$ to zero.

$$\begin{aligned}
&\delta u=0:\frac{\partial N_x}{\partial x}=I_1\frac{\partial^2 u_0}{\partial t^2}+I_2\frac{\partial^2\phi}{\partial t^2}\\
&\delta\boldsymbol{w}=\mathbf{0}:\frac{\partial\boldsymbol{Q}_{\boldsymbol{x}}}{\partial\boldsymbol{x}}+\frac{\partial}{\partial\boldsymbol{x}}\left(\boldsymbol{N}_{\boldsymbol{x}_{\mathbf{0}}}\frac{\partial\boldsymbol{w}_{\mathbf{0}}}{\partial\boldsymbol{x}^{\mathbf{2}}}\right)=\boldsymbol{I}_{\mathbf{1}}\frac{\partial^{\mathbf{2}}\boldsymbol{w}_{\mathbf{0}}}{\partial\boldsymbol{t}^{\mathbf{2}}}\\
&\delta\phi=0:\frac{\partial M_x}{\partial x}-Q_X=I_2\frac{\partial^2 u_0}{\partial t^2}+I_3\frac{\partial^2\phi}{\partial t^2}
\end{aligned} \tag{24}$$

N_x and M_x are defined as follows:

$$\begin{aligned}
&\boldsymbol{N}_{\boldsymbol{x}}=\overline{\boldsymbol{N}}_{\boldsymbol{x}}+\boldsymbol{N}_{\boldsymbol{x}}^{\boldsymbol{T}}\\
&\boldsymbol{M}_{\boldsymbol{x}}=\overline{\boldsymbol{M}}_{\boldsymbol{x}}+\boldsymbol{M}_{\boldsymbol{x}}^{\boldsymbol{T}}
\end{aligned} \tag{25}$$

In the relationships above, N_x, M_x, N^T, M^T, and Q_x are the axial load, bending moment, axial force arising from thermal stress, thermal bending moment, and shear force, respectively.

$$\begin{Bmatrix} \overline{N_x} \\ \overline{M_x} \\ \overline{Q_x} \end{Bmatrix} = \int_{-h/2}^{h/2} \begin{Bmatrix} \sigma_{xx} \\ z\sigma_{xx} \\ \tau_{xz} \end{Bmatrix} dz$$
$$\begin{Bmatrix} N_x^T \\ M_x^T \\ Q_x^T \end{Bmatrix} = \int_{-h/2}^{h/2} \begin{Bmatrix} \alpha Q_{11} \\ \alpha z Q_{11} \\ 0 \end{Bmatrix} \Delta T dz \tag{26}$$

Where Q_x^T is the shear force arising from the thermal stress and is usually equal to zero. Also

$$\Delta T(z) = T(z) - T_{Ref} \tag{27}$$

In this study, the bottom surface temperature of the beam has been ascribed to the reference temperature, i.e., $T_{Ref} = T_b$. Thus, $\overline{N}_x$, $\overline{M}_x$, and Q_X can be defined as follows:

$$\overline{N}_x = A_{11}\frac{\partial u_0}{\partial x} + B_{11}\frac{\partial \phi}{\partial x}$$
$$\overline{M}_x = B_{11}\frac{\partial u_0}{\partial x} + D_{11}\frac{\partial \phi}{\partial x}$$
$$Q_X = KA_{55}\left(\frac{\partial w_0}{\partial x} + \phi\right) \tag{28}$$

In Eq. (28), k denotes the shear modification coefficient, which its amount depends on the cross section and it is taken 5/6 for rectangular cross section. In addition, the stiffness components and inertia terms are calculated in the underlying equations:

$$(A_{11}, B_{11}, D_{11}) = \int_{-h/2}^{h/2} Q_{11}(z)\left(1, z, z^2\right) dz$$
$$A_{55} = \int_{-h/2}^{h/2} Q_{55}(z)\, dz$$
$$(I_1, I_2, I_3) = \int_{-h/2}^{h/2} \rho(z)\left(1, z, z^2\right) dz \tag{29}$$

Substituting Eq. (28) into Eq. (24) leads to the differential equations of motion as follows:

$$A_{11}\frac{\partial^2 u_0}{\partial x^2} + B_{11}\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial N_x^T}{\partial x} = I_1\frac{\partial^2 u_0}{\partial t^2} + I_2\frac{\partial^2 \phi}{\partial t^2}$$
$$KA_{55}\left(\frac{\partial^2 w_0}{\partial x^2} + \frac{\partial \phi}{\partial x}\right) + N_{x_0}\frac{\partial^2 w_0}{\partial x^2} = I_1\frac{\partial^2 w_0}{\partial t^2}$$
$$B_{11}\frac{\partial^2 u_0}{\partial x^2} + D_{11}\frac{\partial^2 \phi}{\partial x^2} - KA_{55}\left(\frac{\partial w_0}{\partial x} + \phi\right) + \frac{\partial M_x^T}{\partial x} = I_2\frac{\partial^2 u_0}{\partial t^2} + I_3\frac{\partial^2 \phi}{\partial t^2} \tag{30}$$

The boundary conditions of the studied beam are assumed to be clamped that is described as:

$$Clamped\,(C): \; u_0 = w_0 = \phi = 0 \tag{31}$$

The following dimensionless quantities make it possible that these studied results will be independent of dimensions.

$$\xi = \frac{x}{L}$$
$$(u, w) = \left(\frac{u_0}{h}, \frac{w_0}{h}\right)$$
$$\eta = \frac{L}{h}$$
$$\tau = \frac{t}{L}\sqrt{\frac{A_{110}}{I_{10}}}$$
$$(\overline{I}_1, \overline{I}_2, \overline{I}_3) = \left(\frac{I_1}{I_{10}}, \frac{I_2}{I_{10}h}, \frac{I_3}{I_{10}h^2}\right) \tag{32}$$

$$(a_{11}, a_{55}, b_{11}, d_{11}) = \left(\frac{A_{11}}{A_{110}}, \frac{A_{55}}{A_{110}}, \frac{B_{11}}{A_{110}h}, \frac{D_{11}}{A_{110}h^2}\right)$$
$$(N_x^*, \lambda_{x_0}) = \frac{(N_x^T, N_{x_0})}{A_{110}}$$
$$M_x^* = \frac{M_x^T h}{D_{110}}$$
$$\omega = \Omega L\sqrt{\frac{I_{10}}{A_{110}}}$$
$$\left(\bar{a}_{11}, \bar{a}_{55}, \bar{b}_{11}, \bar{d}_{11}\right) = \left(\frac{A_{11}h^2}{D_{110}}, \frac{A_{55}h^2}{D_{110}}, \frac{B_{11}h}{D_{110}}, \frac{D_{11}}{D_{110}}\right)$$
$$\gamma = \frac{A_{110}h_0^2}{D_{110}}$$

In the equations above, A_{110}, D_{110}, and I_{10} are the same as A_{11}, D_{11}, I_1 in the pure metal beam without porosity and nanofillers. According to Eq. (32), the dimensionless form of governing Eq. (30) is rewritten as:

$$\begin{aligned}
&a_{11}\frac{\partial^2 u}{\partial \xi^2} + b_{11}\frac{\partial^2 \phi}{\partial \xi^2} + \eta\frac{\partial N_x^*}{\partial \xi} = \bar{I}_1\frac{\partial^2 u}{\partial \tau^2} + \bar{I}_2\frac{\partial^2 \phi}{\partial \tau^2}\\
&Ka_{55}\left(\frac{\partial^2 w}{\partial \xi^2} + \eta\frac{\partial \phi}{\partial \xi}\right) + \lambda_{x_0}\frac{\partial^2 w}{\partial \xi^2} = \bar{I}_1\frac{\partial^2 w}{\partial \tau^2}\\
&\bar{b}_{11}\frac{\partial^2 u}{\partial \xi^2} + \bar{d}_{11}\frac{\partial^2 \phi}{\partial \xi^2} - K\bar{a}_{55}\eta\left(\frac{\partial w}{\partial \xi} + \eta\phi\right) + \eta\frac{\partial M_x^*}{\partial \xi} = \gamma\left(\bar{I}_2\frac{\partial^2 u}{\partial \tau^2} + \bar{I}_3\frac{\partial^2 \phi}{\partial \tau^2}\right)
\end{aligned} \tag{33}$$

In the same way, the boundary conditions can be dimensionless.

$$\textit{Clamped}\,(C): \quad u = w = \phi = 0 \qquad \xi = 0 \text{ or } 1 \tag{34}$$

It is assumed that the sum of static and dynamic displacement field leads to the total displacement field.

$$\begin{aligned}
&u = u_s + u_q\\
&w = w_s + w_q\\
&\phi = \phi_s + \phi_q
\end{aligned} \tag{35}$$

In Eq. (35), subscripts ‘s’ and ‘q’ refer to the points in the static and dynamic field, respectively.
The primitive stress originating from the stable thermal field can be calculated by solving static equations.

$$\begin{aligned}
&a_{11}\frac{\partial^2 u_s}{\partial \xi^2} + b_{11}\frac{\partial^2 \phi_s}{\partial \xi^2} + \eta\frac{\partial N_x^*}{\partial \xi} = 0\\
&Ka_{55}\left(\frac{\partial^2 w_s}{\partial \xi^2} + \eta\frac{\partial \phi_s}{\partial \xi}\right) = 0\\
&\bar{b}_{11}\frac{\partial^2 u_s}{\partial \xi^2} + \bar{d}_{11}\frac{\partial^2 \phi_s}{\partial \xi^2} - K\bar{a}_{55}\eta\left(\frac{\partial w_s}{\partial \xi} + \eta\phi_s\right) + \eta\frac{\partial M_x^*}{\partial \xi} = 0
\end{aligned} \tag{36}$$

Obviously, Eq. (36) is obtained by dropping the inertia terms and λ_{x_0} in Eq. (33). The boundary conditions are those mentioned already. According to Eqs. (33) and (35), the dynamic field equations will be as follows:

$$\begin{aligned}
&a_{11}\frac{\partial^2 u_q}{\partial \xi^2} + b_{11}\frac{\partial^2 \phi_q}{\partial \xi^2} = \bar{I}_1\frac{\partial^2 u_q}{\partial \tau^2} + \bar{I}_2\frac{\partial^2 \phi_q}{\partial \tau^2}\\
&Ka_{55}\left(\frac{\partial^2 w_q}{\partial \xi^2} + \eta\frac{\partial \phi_q}{\partial \xi}\right) + \lambda_{x_0}\frac{\partial^2 w_q}{\partial \xi^2} = \bar{I}_1\frac{\partial^2 w_q}{\partial \tau^2}\\
&\bar{b}_{11}\frac{\partial^2 u_q}{\partial \xi^2} + \bar{d}_{11}\frac{\partial^2 \phi_q}{\partial \xi^2} - K\bar{a}_{55}\eta\left(\frac{\partial w_q}{\partial \xi} + \eta\phi_q\right) = \gamma\left(\bar{I}_2\frac{\partial^2 u_q}{\partial \tau^2} + \bar{I}_3\frac{\partial^2 \phi_q}{\partial \tau^2}\right)
\end{aligned} \tag{37}$$

The boundary conditions in the dynamic analysis become homogeneous.

$$\textit{Clamped}\,(C): \quad u_q = w_q = \phi_q = 0, \qquad \xi = 0, 1 \tag{38}$$

3.2. Thermal field

The one-dimensional thermal gradient affecting the beam, before undergoing dynamic deformation, is along the thickness direction. It is considered that the top and bottom beam surfaces are held at temperatures T_u and T_b, respectively. Therefore, the thermal boundary conditions are:

$$\begin{aligned} &T = T_b \quad \rightarrow \quad z = -h/2 \\ &T = T_u \quad \rightarrow \quad z = h/2 \end{aligned} \tag{39}$$

The temperature profile changes linearly along the thickness direction as follows:

$$T\left(z\right) = T_b + \Delta T_{tot}\left(\frac{z}{h} + \frac{1}{2}\right) \tag{40}$$

Where

$$\Delta T_{tot} = T_u - T_b \tag{41}$$

3.3. Generalized differential quadrature method (GDQM)

In recent studies, the generalized differential quadrature method (GDQM) is applied as an accurate, efficient and simple numerical technique for structure analysis.

By applying GDQM, the discretization process of the present beam equations is conducted; in fact, this method provides the possibility of solving the present beam equations with high accuracy in a short time.

By GDQM, the rth-order partial derivative of a continuous function f(ξ) with respect to ξ at a given point ξ_i can be estimated as a linear sum of the weighted function values at all of the discrete points in the domain of ξ [34,35].

$$\frac{\partial^n f(\xi_i)}{\partial \xi^n} = \sum_{K=1}^{N} c_{iK}^{(n)} f\left(\xi_K\right), \quad i = 1, \ldots, N,\ n = 1, \ldots, N-1 \tag{42}$$

where N and f(ξ_k) signify the number of sampling points in the axial direction of the beam and functional value at a sample point, respectively. Also, ξ_k, and $c_{ik}^{(r)}$ represent the weighting coefficients of the rth-order derivative. For the first derivative (i.e. r = 1), the weighting coefficients are defined as:

$$C_{ij}^{(1)} = \begin{cases} \dfrac{M^{(1)}(x_i)}{(x_i - x_j)M^{(1)}(x_j)} & i \neq j,\ i = 1, 2, \ldots,\ N \\ -\displaystyle\sum_{j=1, i\neq j}^{N} C_{ij}^{(1)} & i = j,\ i = 1, 2, \ldots, N \end{cases} \tag{43}$$

where

$$M\left(x_i\right) = \prod_{j=1, i\neq j}^{N} \left(x_i - x_j\right) \tag{44}$$

For higher-order derivatives

$$C_{ij}^{(n)} = \begin{cases} n\left[C_{ii}^{(n-1)} C_{ij}^{(1)} - \dfrac{C_{ij}^{(n-1)}}{(x_i - x_j)}\right] & i \neq j,\ i = 1, 2, \ldots,\ N,\ n = 2, 3, \ldots,\ N-1 \\ -\displaystyle\sum_{j=1, i\neq j}^{N} C_{ij}^{(n)} & i = j,\ i = 1, 2, \ldots,\ N \end{cases} \tag{45}$$

Herein, the grid points are dispersed non-uniformly and are given by the following equation:

$$\xi_i = \frac{1}{2}\left(1 - \cos\left(\frac{i-1}{N-1}\pi\right)\right) \quad i = 1, 2, \ldots, N \tag{46}$$

3.4. Thermal buckling analysis

The porous nanocomposite beam is subjected to the thermal field. It is known that the thermal field generates the buckling factor, that is, critical thermal gradient $\Delta T_{tot,cr}$. Therefore, the equation of the thermal buckling is derived by dropping the inertia terms and $\lambda_{x_0} = -N_{x,cr}^{*}$ in the dynamic field equations,

$$\begin{aligned}
&a_{11}\frac{\partial^2 \mathrm{u_q}}{\partial \xi^2}+\mathrm{b}_{11}\frac{\partial^2 \phi_\mathrm{q}}{\partial \xi^2}=0\\
&\mathbf{Ka_{55}}\left(\frac{\partial^2 \mathbf{w_q}}{\partial \xi^2}+\eta\frac{\partial \phi_\mathbf{q}}{\partial \xi}\right)-\boldsymbol{N^*_{x,cr}}\frac{\partial^2 \mathbf{w_q}}{\partial \xi^2}=\mathbf{0}\\
&\bar{\mathbf{b}}_{11}\frac{\partial^2 \mathbf{u_q}}{\partial \xi^2}+\bar{\mathbf{d}}_{11}\frac{\partial^2 \phi_\mathbf{q}}{\partial \xi^2}-\mathbf{K\bar{a}_{55}}\eta\left(\frac{\partial \mathbf{w_q}}{\partial \xi}+\eta\phi_\mathbf{q}\right)=\mathbf{0}
\end{aligned}\tag{47}$$

By applying GDQM expressed in Eqs. (42)–(46), the following relationships are obtained:

$$\begin{aligned}
&\mathrm{a_{11}B_{i1}u_1+a_{11}B_{iN}u_N+a_{11}\sum_{k=2}^{N-1}B_{ik}u_k+b_{11}B_{i1}\phi_1+b_{11}B_{iN}\phi_N+b_{11}\sum_{k=2}^{N-1}B_{ik}\phi_N=0}\\
&ka_{55}\left(B_{i1}w_1+B_{iN}w_N+\sum_{k=2}^{N-1}B_{ik}w_k+\eta A_{i1}\phi_1+\eta A_{iN}\phi_N+\eta\sum_{k=2}^{N-1}A_{ik}\phi_k\right)-N^*_{x,cr}B_{i1}w_1-N^*_{x,cr}B_{iN}w_N-N^*_{x,cr}\sum_{k=2}^{N-1}B_{ik}w_k=0\\
&\bar{b}_{11}B_{i1}u_1+\bar{b}_{11}B_{iN}u_N+\bar{b}_{11}\sum_{k=2}^{N-1}B_{ik}u_k-k\bar{a}_{55}\eta(B_{ik}w_1+B_{iN}w_N+\sum_{k=2}^{N-1}B_{ik}w_k+\eta\phi_i)=0
\end{aligned}\tag{48}$$

where A_{ij} and B_{ij} are the first and second order GDQM weighting coefficients, respectively. The associated boundary conditions are discreted in the same way. The boundary conditions equations are:

$$\mathbf{C-C\ beam}:\begin{cases}\boldsymbol{u_1=0}\\ \boldsymbol{w_1=0}\\ \boldsymbol{\phi_1=0}\\ \boldsymbol{u_N=0}\\ \boldsymbol{w_N=0}\\ \boldsymbol{\phi_N=0}\end{cases}\tag{49}$$

Implementing the boundary conditions into Eq. (50) gives the algebraic set as:

$$\begin{bmatrix}[S_{bb}] & [S_{bd}]\\ [S_{db}] & [S_{dd}]\end{bmatrix}\begin{Bmatrix}\{U_b\}\\ \{U_d\}\end{Bmatrix}=\Delta T_{tot,cr}\begin{bmatrix}[0] & [0]\\ [A_{db}] & [A_{dd}]\end{bmatrix}\begin{Bmatrix}\{U_b\}\\ \{U_d\}\end{Bmatrix}\tag{50}$$

where $\{U_b\}$ and $\{U_d\}$ are mentioned in the following equations:

$$\{\boldsymbol{U_b}\}=\left\{\{\boldsymbol{u_b}\},\{\boldsymbol{w_b}\},\{\boldsymbol{\phi_b}\}\right\}^{\boldsymbol{T}}\tag{51}$$

and

$$\{\boldsymbol{U_d}\}=\left\{\{\boldsymbol{u_d}\},\{\boldsymbol{w_d}\},\{\boldsymbol{\phi_d}\}\right\}^{\boldsymbol{T}}\tag{52}$$

In Eq. (54), subscripts ‘b’ and ‘d’ refer to the points on the boundary and in the interior domain, respectively, and $\bar{I}_i$ is dimensionless inertia terms matrices. Eliminating the boundary degrees of freedom provides the underlying equation:

$$\left([S]-\Delta T_{tot,cr}[A]\right)\{u_d\}=\{0\}\tag{53}$$

where

$$\begin{aligned}
&[S]=[S_{dd}]-[S_{db}][S_{bb}]^{-1}[S_{bd}]\\
&[A]=[A_{dd}]-[A_{db}][S_{bb}]^{-1}[S_{bd}]
\end{aligned}\tag{54}$$

The critical thermal gradient - $\Delta T_{cr,tot}$ - of the beam is found by solving the generalized eigenvalue problem Eq. (53). Clearly, the square of the lowest positive eigenvalue of Eq. (53) gives the critical thermal gradient of the considered beam.

4. Numerical results and discussion

To investigate the effects of GPLs and porosity on the thermal buckling of the nanocomposite beams, a parametric study is done in detail. The main parameters assessed in this paper are the dispersion patterns, weight fraction (wt.%), geometry, and size of GPLs as well as the porosity coefficient and distribution. Aluminum Al, magnesium Mg, copper Cu, nickel Ni, and titanium Ti play the role of material matrix in this study. According to [36–40], the properties of these materials are tabulated in Table 1. In addition, the material property and geometry parameters of GPLs are: $w_{GPL}=1.5\ (\mu m)$, $l_{GPL}=2.5\ (\mu m)$, $t_{GPL}=1.5\ (nm)$, $E_{GPL}=1.01\ (TPa)$, $\rho_{GPL}=1062.5kg/m^3$, $\nu_{GPL}=0.186$, $\alpha_{GPL}=-3.75\times10^{-6}\ k^{-1}$ [14,41,42].

Table 1
Typical material properties of metals.

	Density (kg/m^3)	Poisson's ratio	Young's modulus (GPa)	Thermal expansion coefficient (k^{-1})
Aluminum	2689.8	0.34	68.3	23×10^{-6}
Copper	8960	0.34	130	17×10^{-6}
Nickel	8908	0.31	210	13×10^{-6}
Magnesium	1740	0.35	45	25×10^{-6}
Titanium	4506	0.33	116	8.6×10^{-6}

4.1. Validation and convergence study

Kitipornchai et al. [43] have investigated the free vibration and elastic buckling of functionally graded porous nanocomposite beams. Any solution for the present problem is not available, thus the results of this study for $\Delta T_{tot}=0$ were compared with those of reference [43] to validate our results. By assuming $\Delta T_{tot}=0$, the equations of motion (Eq. (30) are expressed as follows:

$$\begin{aligned}
&A_{11}\frac{\partial^2 u_0}{\partial x^2}+B_{11}\frac{\partial^2\phi}{\partial x^2}=I_1\frac{\partial^2 u_0}{\partial t^2}+I_2\frac{\partial^2\phi}{\partial t^2}\\
&KA_{55}\left(\frac{\partial^2 w_0}{\partial x^2}+\frac{\partial\phi}{\partial x}\right)+N_{x_0}\frac{\partial^2 w_0}{\partial x^2}=I_1\frac{\partial^2 w_0}{\partial t^2}\\
&B_{11}\frac{\partial^2 u_0}{\partial x^2}+D_{11}\frac{\partial^2\phi}{\partial x^2}-KA_{55}\left(\frac{\partial w_0}{\partial x}+\phi\right)=I_2\frac{\partial^2_{u_0}}{\partial t^2}+I_3\frac{\partial^2\phi}{\partial t^2}
\end{aligned}\tag{55}$$

and the set of the above equations become dimensionless as follows:

$$\begin{aligned}
&\mathbf{a}_{11}\frac{\partial^2\mathbf{u}}{\partial\xi^2}+\mathbf{b}_{11}\frac{\partial^2\phi}{\partial\xi^2}=\bar{\mathrm{I}}_1\frac{\partial^2\mathbf{u}}{\partial\tau^2}+\bar{\mathrm{I}}_2\frac{\partial^2\phi}{\partial\tau^2}\\
&\mathbf{Ka}_{55}\left(\frac{\partial^2\mathbf{w}}{\partial\xi^2}+\eta\frac{\partial\phi}{\partial\xi}\right)+\lambda_{\mathbf{x}_0}\frac{\partial^2\mathbf{w}}{\partial\xi^2}=\bar{\mathrm{I}}_1\frac{\partial^2\mathbf{w}}{\partial\tau^2}\\
&\mathbf{b}_{11}\frac{\partial^2\mathbf{u}}{\partial\xi^2}+\mathbf{d}_{11}\frac{\partial^2\phi}{\partial\xi^2}-\mathbf{Ka}_{55}\eta\left(\frac{\partial\mathbf{w}}{\partial\xi}+\eta\phi\right)=\bar{\mathrm{I}}_2\frac{\partial^2_{\mathbf{u}}}{\partial\tau^2}+\bar{\mathrm{I}}_3\frac{\partial^2\phi}{\partial\tau^2}
\end{aligned}\tag{56}$$

To determine the dimensionless fundamental frequency, the assumption of $\lambda_{x_0}=0$ is applied in Eq. (56) and it is obtained as follows:

$$\begin{bmatrix}[S_{bb}] & [S_{bd}]\\ [S_{db}] & [S_{dd}]\end{bmatrix}\begin{Bmatrix}\{U_b\}\\ \{U_d\}\end{Bmatrix}=\omega^2\begin{bmatrix}[\mathbf{0}] & [\mathbf{0}]\\ [\mathbf{0}] & [\bar{I}_i]\end{bmatrix}\begin{Bmatrix}\{U_b\}\\ \{U_d\}\end{Bmatrix}\tag{57}$$

By solving the generalized eigenvalue problem concluded from Eq. (57), the frequency has been calculated.

To investigate the buckling, the following assumptions have been applied:

$$\lambda_{\mathbf{x}_0}=-\mathbf{P},\ \omega=\mathbf{0}\tag{58}$$

Where P denotes the critical buckling load.

Implementing the above assumptions into Eq. (56) leads to the following algebraic system:

$$\begin{bmatrix}[S_{bb}] & [S_{bd}]\\ [S_{db}] & [S_{dd}]\end{bmatrix}\begin{Bmatrix}\{U_b\}\\ \{U_d\}\end{Bmatrix}=P\begin{bmatrix}[\mathbf{0}] & [\mathbf{0}]\\ [A_{db}] & [A_{dd}]\end{bmatrix}\begin{Bmatrix}\{U_b\}\\ \{U_d\}\end{Bmatrix}\tag{59}$$

Similar to the analysis of the free vibration, the critical buckling load is obtained through solving the concluded algebraic system. The dimensionless critical buckling load and fundamental frequency gained in the present analysis are compared with the similar results given by [43] in Tables 2 and 3.

One can see, our results are in accord to the previously reported ones with the differences in the range of 0–0.36%.

Fig. 3 depicts result convergence. $\Delta T_{cr,tot}$ is organized by polynomial item number N. In N=7, results, in all cases, convergence to a certain amount. As seen, the GDQ method increases the solving convergence rate. Since time is of the essence in numerical study, thus GDQ method will be proper choice for such studies.

4.2. Analysis of thermal buckling

The Table 4 that comprises four categories of porosity coefficient illustrates $\Delta T_{cr,tot}$ of the beam with the various numbers of layers. With regard to multilayer structure of the present beam, to reduce cost and improve accuracy, either the number of beam layer should be kept to minimum or the results should be close to the results of continuous one. It has been supposed that n=1000 represents the continues beam. As noticed, fourteenth layer beam meets the aforementioned points.

The reinforcing effect of GPLs on the five various beams – Al, Mg, Ni, Cu, and Ti beams – in two weight fractions is highlighted in Fig. 4. In general, introducing GPLs to the beams shots up $\Delta T_{cr,tot}$ except Nickel beam. Regarding the high Young's modules of Nickel,

Table 2
Dimensionless critical buckling of porous beams (porosity distribution 1, GPL pattern A, $l/h = 20$, C-C copper-matrix beam).

Λ_{GPL}	n	$e_0 = 0$		$e_0 = 0.2$		$e_0 = 0.4$		$e_0 = 0.6$	
		Present study	[43]	Present study	[43]	Present study	[43]	Present study	[43]
1 wt.%	2	0.008526	0.008550	0.007215	0.007218	0.005924	0.005917	0.004650	0.004647
	6	0.014274	0.014323	0.013049	0.013057	0.011794	0.011784	0.010496	0.010486
	10	0.014746	0.014798	0.013567	0.013572	0.012347	0.012333	0.011067	0.011063
	14	0.014875	0.014929	0.013705	0.013714	0.012503	0.012486	0.011232	0.011224
	18	0.014931	0.014982	0.013760	0.013773	0.012568	0.012549	0.011296	0.011290
	10000	0.015014	0.015065	0.013853	0.013863	0.012660	0.012645	0.011397	0.011392
0 wt.%	14	0.007946	0.007946	0.007313	0.007316	0.006696	0.006693	0.006076	0.006076
	10000	0.007983	0.007986	0.007359	0.007362	0.006742	0.006745	0.006132	0.006135

Table 3
Dimensionless frequency of porous beams (porosity distribution 1, GPL pattern A, $\boldsymbol{l/h = 20}$, C-C copper-matrix beam).

Λ_{GPL}	n	$e_0 = 0$		$e_0 = 0.2$		$e_0 = 0.4$		$e_0 = 0.6$	
		Present study	[43]	Present study	[43]	Present study	[43]	Present study	[43]
1 wt.%	2	0.3371	0.3376	0.3217	0.3217	0.3044	0.3042	0.2846	0.2845
	6	0.4381	0.4390	0.4334	0.4336	0.4291	0.4289	0.4261	0.4259
	10	0.4455	0.4464	0.4420	0.4421	0.4390	0.4388	0.4373	0.4372
	14	0.4475	0.4484	0.4442	0.4444	0.4418	0.4415	0.4405	0.4403
	18	0.4483	0.4492	0.4451	0.4454	0.4429	0.4426	0.4417	0.4416
	10000	0.4496	0.4505	0.4466	0.4468	0.4445	0.4442	0.4437	0.4436
0 wt.%	14	0.3159	0.3159	0.3134	0.3134	0.3122	0.3121	0.3128	0.3128
	10000	0.3166	0.3167	0.3143	0.3144	0.3132	0.3132	0.3142	0.3142

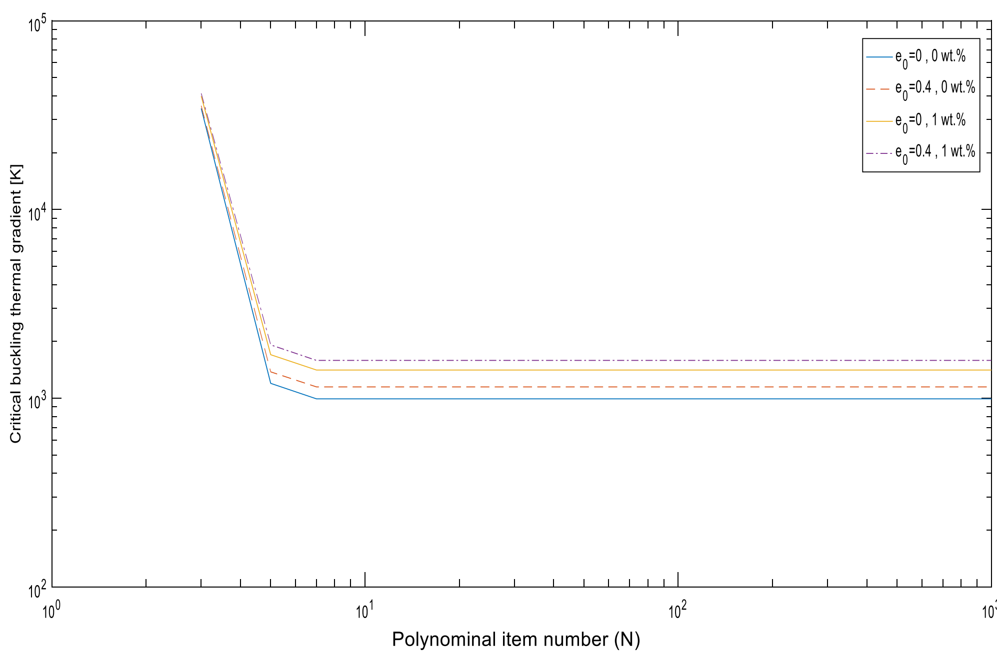

Fig. 3. Critical buckling thermal gradient of nanocomposite beams (porosity distribution 1, GPL pattern A, l/h=20, C-C copper-matrix beam).

this exception is justifiable. On the contrary, in respect of the high thermal expansion coefficient, as well as the low Young's modules of Magnesium, the most increase belongs to Mg beam. Because of inherent limitation of Mg such as high reactivity, the Copper beam that had respectable performance was studied in the following discussion.

Fig. 5 shows the role of GPLs dispersion pattern in the enhancement of beam stiffness. As observed, in the beams with GPLs pattern A and B, introducing a little amount of GPLs increases $\Delta T_{cr,tot}$ dramatically. To conclude, symmetry of GPLs distribution patterns is ultra-efficient factor for the stiffness of the present structure. On the other hand, asymmetric pattern B has adverse effect on its. Meanwhile, according to Fig. 5 and Table 5, in a certain amount of GPLs weight fraction, as porosity coefficient increases the $\Delta T_{cr,tot}$ increment decreases. In the beams with GPLs pattern A, this trend is more obvious than that of B and C.

Fig. 6 displays the changing trend in the $\Delta T_{cr,tot}$ of beams with various GPLs pattern and porosity distribution. As noticed, regardless of the porosity distribution of the beams, the trends of the beams are similar to the Fig. 5. It is evident that the effect of the GPLs pattern

Table 4
Critical thermal gradient of porous beams (porosity distribution 1, GPL pattern A, l/h=20, C-C copper-matrix beam).

Λ_{GPL}	n	$e_0 = 0$	$e_0 = 0.2$	$e_0 = 0.4$	$e_0 = 0.6$
1 wt.%	2	817.4158	816.7211	814.9772	811.7689
	6	1345.1046	1410.4462	1490.7189	1594.9471
	10	1387.7248	1461.9159	1552.3115	1667.3080
	14	1399.4976	1475.6047	1569.7340	1688.2954
	18	1404.4486	1481.0781	1576.8685	1696.3517
	1000	1411.8653	1490.2281	1587.0377	1709.1116
0 wt.%	14	993.1875	1056.7240	1141.6566	1257.2135
	1000	993.1875	1069.0089	1148.6644	1266.8268

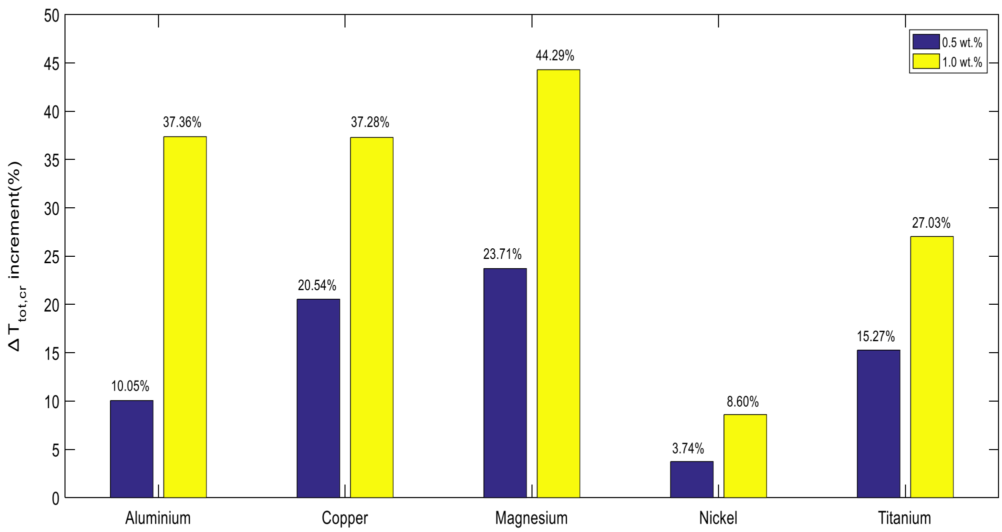

Fig. 4. Effect of GPLs on the $\Delta T_{cr,tot}$ change of porous beams versus different metal matrixes (porosity distribution 1, GPL pattern A, e_0=0.5, l/h=20, C-C copper-matrix beam).

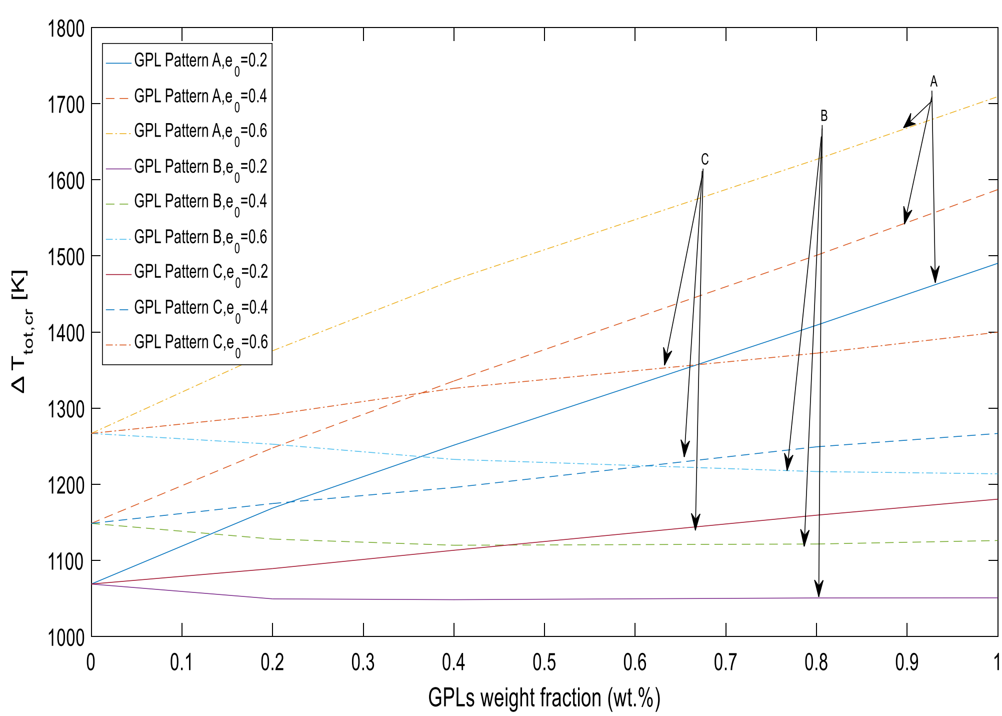

Fig. 5. Effect of GPLs pattern and porosity coefficient on $\Delta T_{cr,tot}$ of nanocomposite beams (porosity distribution 1, l/h=20, C-C copper-matrix beam, ΔT_{tot}=250 [K]).

Table 5
Effect of GPLs pattern and porosity coefficient on the $\boldsymbol{\Delta T_{cr,\,tot}}$ variations of nanocomposite beams (porosity distribution 1, l/h=20, C-C copper-matrix beam).

Λ_{GPL} (wt.%)	$(\Delta T_{tot,cr})$ increment (%) $e_0 = 0.2$	$(\Delta T_{tot,cr})$ increment (%) $e_0 = 0.4$	$(\Delta T_{tot,cr})$ increment (%) $e_0 = 0.6$
0	0.00	0.00	0.00
0.2	9.33	8.63	8.57
0.4	17.06	16.28	15.92
0.8	31.80	30.65	28.40
1	39.40	38.16	34.91

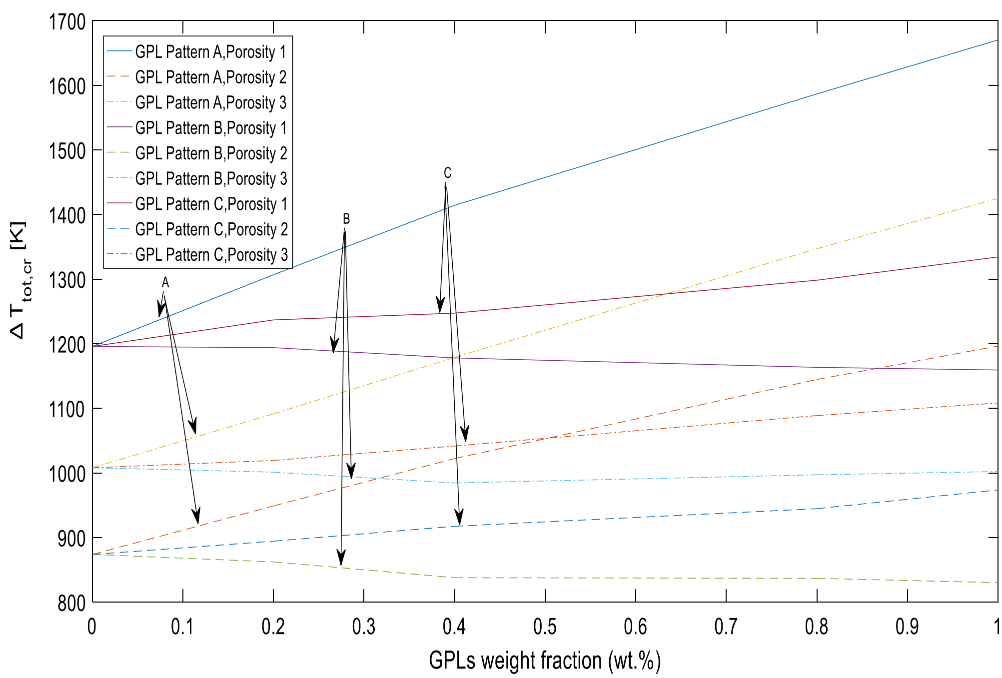

Fig. 6. Effect of GPL pattern and porosity distribution on $\boldsymbol{\Delta T_{cr,\,tot}}$ (e_0=0.5, l/h=20, C-C copper-matrix beam).

Table 6
Effect of GPL pattern A and porosity distributions on the $\boldsymbol{\Delta T_{cr,tot}}$ variations (e_0=0.5, l/h=20, C-C copper-matrix beam).

Λ_{GPL} (wt.%)	$(\Delta T_{tot,cr})$ change (%) Porosity (distribution 1)	$(\Delta T_{tot,cr})$ change (%) Porosity (distribution 2)	$(\Delta T_{tot,cr})$ change (%) Porosity (distribution 3)
0	0.00	0.00	0.00
0.2	9.28	8.60	8.27
0.4	18.27	17.00	16.94
0.8	32.69	30.99	33.64
1	39.68	36.92	41.37

is far more than the porosity distribution. Finally, $\Delta T_{cr,\,tot}$ and consequently stiffness coming from beam with porosity distribution 1 and GPLs pattern A, are the highest.

Tables 6 and 7 compare the increase rate of the critical thermal gradient in the nanocomposite beams reinforced by the symmetric GPLs patterns. As observed, the increase rate of the beams reinforced by GPLs pattern A is higher than that of C.

Fig. 7 outlines information about the effect of the GPLs' geometry on the thermal buckling of the studied beams. To achieve this goal, the effect of l_{GPLs}/t_{GPLs} and l_{GPLs}/w_{GPLs} - assuming constant l_{GPLs} - on $\Delta T_{cr,\,tot}$ is considered. A glance at the graph reveals that at first, in all cases, $\Delta T_{cr,\,tot}$ reaches a pick amounting, next lines get a steady trend approximately, and lastly at $l_{GPLs}/t_{GPLs} = 10^4$, $\Delta T_{cr,\,tot}$ orients to around $1640^\circ K$ which is independent of l_{GPLs}/w_{GPLs}. As shown in the illustration, as l_{GPLs}/t_{GPLs} rises, the $\Delta T_{cr,\,tot}$ will be enhanced. Thereby, it can be concluded that GPLs with less single graphene layer perform better.

$\Delta T_{cr,\,tot}$ values of porous nanocomposite beams reinforced by GPLs are listed for the different slenderness ratios in Table 8. Clearly, $\Delta T_{cr,\,tot}$ falls with a rise in the slenderness ratio.

Table 7
Effect of GPL pattern C and porosity distributions on the $\boldsymbol{\Delta T_{cr,tot}}$ variations (e_0=0.5, l/h=20, C-C copper-matrix beam).

Λ_{GPL} (wt.%)	$(\Delta T_{tot,cr})$ change (%) Porosity (distribution 1)	$(\Delta T_{tot,cr})$ change (%) Porosity (distribution 2)	$(\Delta T_{tot,cr})$ change (%) Porosity (distribution 3)
0	0.00	0.00	0.00
0.2	3.43	2.35	1.12
0.4	4.31	4.99	3.34
0.8	8.57	8.11	8.01
1	11.58	11.44	9.95

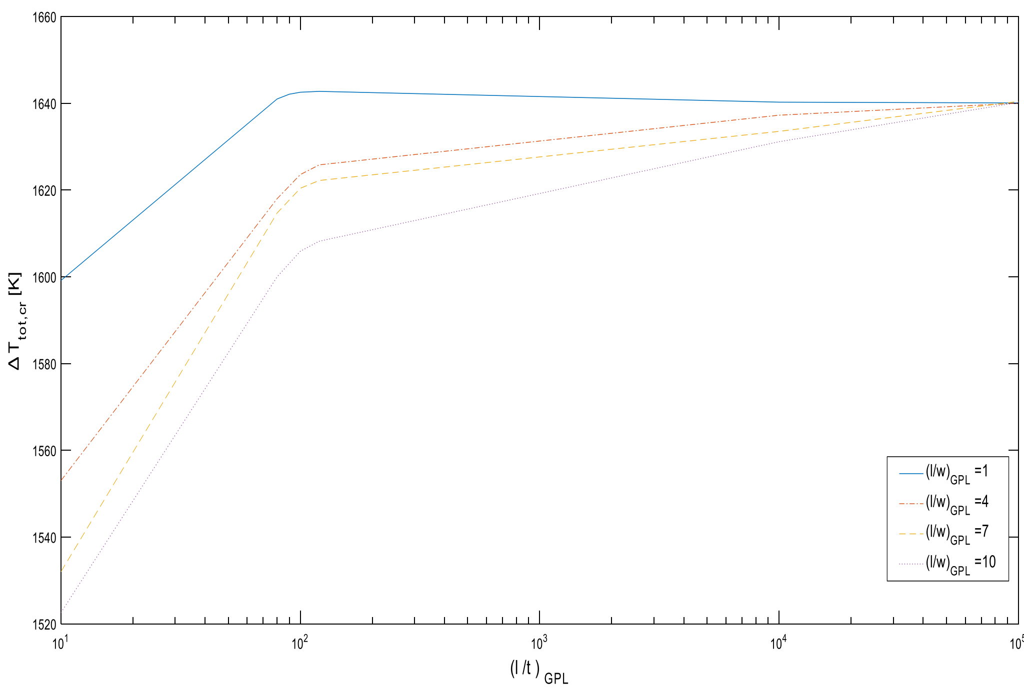

Fig. 7. Effect of size and geometry of GPLs on $\boldsymbol{\Delta T_{cr,tot}}$ of nanocomposite beams (GPL pattern A, porosity distribution 1, e_0=0.5, l/h=20, Λ_{GPL}=1 wt.%, C-C copper–matrix beam).

Table 8
Critical thermal gradient in beams with various slenderness ratios (GPL pattern A, porosity distribution 1, e_0=0.5, C-C copper-matrix beam).

GPL weight fraction (wt.%)	0.0	0.2	0.4	0.6	0.8	1.0
$L/h = 20$	1195.7476	1306.7680	1414.1605	1479.4000	1586.6614	1670.1911
$L/h = 30$	532.8238	583.9015	624.9700	665.7689	687.6889	734.5603
$L/h = 40$	330.7304	341.5417	365.1141	373.8038	398.6900	423.6767

5. Conclusions

The thermal buckling of the multilayer FG porous nanocomposite beams reinforced by GPLs was numerically analyzed based on GDQM. Employing the Timoshenko beam theory and Hamilton principle leads to derive the equations of motion. The boundary conditions of the beam were taken to be clamped. The nanocomposite multilayer beam was subjected to the linear thermal gradient. The effects of porosity and GPLs on the thermal buckling of the beams were presented. It is concluded that:

1- The GDQ method heightens the rate of solving process strikingly.
2- Beam stiffness is noticeably affected by GPLs dispersion pattern:
   a) In the case of the non-uniform symmetric GPLs dispersion pattern A and the uniform GPLs dispersion pattern C introducing a little amount of nanofiller soars the $\Delta T_{cr,tot}$.
   b) Asymmetric one – the GPLs dispersion pattern B - results in a downward tendency negligibly.
3- Either GPLs dispersion pattern or porosity distribution is effective on studied beam behavior but the effect of nanofiller pattern is predominant. Therefore, the determinative factor of beam stiffness trend is that stronger one.
4- In certain amount of GPLs, the $\Delta T_{cr,tot}$ increment and porosity coefficient have an inverse relation.

5- Beam with non-uniform asymmetric porosity distribution and GPLs pattern – porosity distribution 1 and GPLs pattern A - offers the highest $\Delta T_{cr,\,tot}$, subsequently the most stiffness.
6- In geometric viewpoint, larger surface area and less single graphene layer are the preferred parameters for beam performance improvement.
7- A reduction in beam thickness h causes a collapse in $\Delta T_{cr,\,tot}$.

Considering the aforementioned points, it can be expected that structures like the studied nanocomposite beams have a promising future ahead. It is felt the necessity for further researches to understand more about them.

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Appendix A

In this research, due to the multilayer structure of the beam, the stiffness components (A_{11}, B_{11}, D_{11}, and A_{55}), inertia items (I_1, I_2, and I_3), thermal axial load (N_x^T), and thermal bending moment (M_x^T) of the laminated porous nanocomposite beams can be calculated by:

$$\begin{aligned}
A_{11} &= \frac{h}{n}\sum_{i=1}^{n}\frac{E_1(h\beta)\left[1-e_0\lambda(h\beta)\right]}{1-\nu^2(h\beta)} \\
B_{11} &= \frac{h}{n}\sum_{i=1}^{n}\frac{E_1(h\beta)\left[1-e_0\lambda(h\beta)\right]}{1-\nu^2(h\beta)}(h\beta) \\
D_{11} &= \frac{h}{n}\sum_{i=1}^{n}\frac{E_1(h\beta)\left[1-e_0\lambda(h\beta)\right]}{1-\nu^2(h\beta)}(h\beta)^2 \\
A_{55} &= \frac{h}{n}\sum_{i=1}^{n}\frac{E_1(h\beta)\left[1-e_0\lambda(h\beta)\right]}{2\left[1+\nu(h\beta)\right]} \\
I_1 &= \frac{h}{n}\sum_{i=1}^{n}\rho_1(h\beta)\left[1-e_m\lambda(h\beta)\right] \\
I_2 &= \frac{h}{n}\sum_{i=1}^{n}\rho_1(h\beta)\left[1-e_m\lambda(h\beta)\right](h\beta) \\
I_3 &= \frac{h}{n}\sum_{i=1}^{n}\rho_1(h\beta)\left[1-e_m\lambda(h\beta)\right](h\beta)^2 \\
N_x^T &= \sum_{i=1}^{n}\frac{E_1(h\beta)\left[1-e_0\lambda(h\beta)\right]}{1-\nu^2(h\beta)}\alpha_1(h\beta)\Delta T(h\beta) \\
M_x^T &= \sum_{i=1}^{n}\frac{E_1(h\beta)\left[1-e_0\lambda(h\beta)\right]}{1-\nu^2(h\beta)}\alpha_1(h\beta)\,\Delta T\,(h\beta)\,.(h\beta)
\end{aligned} \tag{A.1}$$

where

$$\beta = \frac{z}{h} = \frac{1}{2}+\frac{1}{2n}-\frac{i}{n} \quad (i=1,2,3,\ldots,n) \tag{A.2}$$

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